Bot or Human? Detection of DeepFake Text with Semantic, Emoji, Sentiment and Linguistic Features

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***Abstract*—Detecting machine-generated text (MGT), also known as Deepfake text, has become increasingly important in Artificial Intelligence (AI) age and social media platforms. With the proliferation of MGT and the potential consequences of its dissemination, there is a pressing need to develop effective methods for distinguishing between MGT and human-written text (HWT). Our research aim has two-fold: firstly, to examine the inherent differences between MGT and HWT on Twitter, and secondly, to develop a classifier specifically designed for MGT detection on the platform. This classifier utilizes contextualized text embeddings as its foundation while considering additional linguistic features, sentiment features, and emoji embeddings. Our experimental results demonstrate that incorporating additional features enhances the model's ability to detect MGT. Combining fine-tuned BERT embeddings with emoji and linguistic features using a multi-layer perceptron classifier achieves the highest accuracy rate of 88.3%. Our analysis reveals distinct characteristics of MGT compared to HWT, including differences in engagement behavior, linguistic patterns, named entities, sentiment expressions, and text perplexity. Our research contributes to the field of MGT detection by offering a comprehensive approach that combines semantic text embeddings with supplementary features. The proposed model provides a significant step forward in addressing the challenge of Deepfake text detection.**

Keywords—Artificial Intelligence; Machine Learning; Data Mining; Deepfake Detection; Feature Engineering

I. INTRODUCTION

Social media platforms have become integral to our daily lives and facilitate the rapid exchange of information and ideas. With the development of sophisticated text generative models and tools like ChatGPT [1], the ability to produce coherent text that closely mimics human writing has become a growing concern. This advancement further heightened the capabilities of bots, particularly in the generation of MGT. Ferrara et al. [2] defined bots as computer algorithms that generate content and interact with humans in online social networks, attempting to imitate and potentially alter human behavior. Deepfake or MGT is generated using artificial intelligence (AI) techniques and can be used for benign and malicious purposes. While there are legitimate applications of Deepfake text, the misuse of this technology can lead to the spread of misinformation, hate speech, and manipulation of public opinion. Therefore, detecting Deepfake text has thus become crucial in maintaining the integrity of online platforms and safeguarding users from deceptive content. Traditional approaches, such as manual evaluation and labeling, are impractical and prone to error [3]. Moreover, existing Deepfake text detection systems, including prominent ones like GPTZero and Open AI's detector [4], are ill-equipped to detect short social media texts, which are prevalent on platforms like Twitter. Exploring novel features, linguistic patterns, and contextual cues specific to social media text holds promise in developing detection techniques [33] that can effectively identify Deepfake text in short social media posts.

Our research objectives are to investigate the inherent distinctions between MGT and HWT on Twitter, focusing on identifying critical linguistic and sentiment characteristics that differentiate the two. Secondly, to develop a robust classifier designed explicitly for detecting MGT on the platform. Our research methodology centers on detecting short Deepfake text samples using the TweepFake dataset introduced by Fagni et al. [5]. While the previous studies primarily focused on tweet semantic text content, we aim to develop a more robust detector by incorporating additional features, including emoji, linguistic cues, and sentiment features derived from the tweet content. We leverage both the capabilities of a neural language model (NLM) based and a feature-based approach for our classification task. To ensure the efficacy of our classifier in discerning MGT from HWT, we conducted an exploratory analysis of various linguistic and sentiment features. The structure of this paper is as follows: Section II presents the literature relevant to this research, Section III explains our research methodology, and finally, we discuss and conclude our research contribution and future work in Sections IV and V.

II. LITERATURE REVIEW

A. Deepfake and Fake News

Recent advancements in AI and Deep Learning (DL) have facilitated the generation of synthetic media with remarkable realism, including images, videos, audio, and text. Deepfakes refer to synthetic media manipulated or entirely generated by AI to depict events that did not occur in reality [5]. With the continuous evolution of natural language processing (NLP) techniques, Deepfakes have expanded into text generation [5], enabling the production of MGT that closely mimics HWT. Consequently, this mimicking capability presents significant challenges distinguishing between AI-generated and human-generated content.

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Two prominent forms of misleading information have emerged in synthetic media: Deepfakes and fake news. While both pose significant challenges regarding detection, trust, and manipulation, they differ in their underlying technologies and modes of dissemination [35]. Deepfake technologies utilize AI and DL algorithms to generate highly realistic synthetic media, including images, videos, audio, and text. Deepfakes can manipulate existing content or develop new content to depict events that did not occur in reality [5]. On the other hand, fake news can be understood as intentional or unintentional counterfeit news, misleading or fabricated information presented as genuine news or factual content [3]. Unlike Deepfakes, which are generated using AI technologies, humans can create fake news to deceive or manipulate public opinion. Fake news can take various forms, including articles, headlines, social media posts, or even used images or videos [7]. The dissemination of fake information is often driven by political, social, or economic motives, aiming to shape narratives, spread misinformation, or generate controversy [6].

B. Natural Language Generation and Transformers

Understanding the text generation process by generative language models is essential for developing an effective bot detection system. The introduction of Transformers, a novel neural network architecture based on a self-attention mechanism developed by Google researchers [8], revolutionized the field of natural language generation (NLG). Transformers have distinct advantages over traditional recurrent neural networks (RNNs) [9], as they can learn long-range dependency information without bias towards recent tokens. Consequently, most state-of-the-art NLG models, such as BERT (Bidirectional Encoder Representations from Transformers) [10], and GPT-2 [11] with the Transformer architecture is an unsupervised pre-trained language model trained on text from popular web pages.

C. Machine-Generated Text (MGT) Detection Methods

Existing MGT detection models can be broadly categorized into three types based on complexity [12]: i) Standard machine-learning classifiers utilize machine learning (ML) or DL algorithms. These classifiers are relatively simple, with few parameters, and can be easily deployed. However, they lack adaptability and require new training when dealing with changing corpora; ii) Zero-shot detection approach [34] rely on the availability of a pre-trained generative model, such as GPT-2 [11] and GPT-3 [13], to analyze outputs from itself or similar models, leveraging the probabilities assigned by the model to strings of text. This approach does not require further training or fine-tuning with supervised detection examples; iii) Fine-tuning-based detection approaches use a pre-trained language model, such as BERT [10] or RoBERTa [14], that is fine-tuned on a specific dataset to achieve higher. Unlike zero-shot detection, this approach requires supervised examples for further training.

There are two types of MGT detection approaches [15]: i) Feature-based approaches use natural language processing (NLP) techniques to extract feature vectors from input sequences, which are then classified using algorithms such as shallow ML models or deep neural networks (DNN). These approaches are also known as standard ML classifier approaches; ii) NLM-based approaches leverage DL architectures, particularly RNNs, and incorporate features derived from state-of-the-art Transformer-based neural language models like GPT (Generative Pre-trained Transformer) which can learn complex patterns and relationships within language data by processing sequential input. Both zero-shot detection [34] and fine-tuning-based detection can be categorized as NLM-based approaches.

D. Related Work

The dataset used in our research is built upon the foundation laid by Fagni et al. [5], which addressed detecting MGTs on social media by introducing the TweepFake dataset. This dataset represents the labeled Twitter dataset comprising exclusively human and machine-generated tweets by manually selecting tweets from genuine human accounts and their corresponding fake bot counterparts operated by individuals on the Twitter platform. This labeling approach ensured that the dataset encompassed tweets generated by text generation algorithms. Table I summarizes and compares related studies in detecting MGT, highlighting their key. While previous works primarily considered tweet semantic text content, our goal is to develop a detector that incorporates additional features, including emoji, linguistic, and sentiment features. By doing so, we aim to leverage the capabilities of both the NLM-based approach and feature-based approach for our classification task. The NLM-based approach enables the incorporation of features derived from Transformer-based models.

TABLE I. PRIOR STUDIES FOR MACHINE-GENERATED TEXT DETECTION

Related Work	Dataset	Feature-based Approach features	NLM
Fagni et al.[5]	TweepFake [5]	Semantic Embeddings	✓
Saravani et al. [16]	TweepFake	Semantic Embeddings	✓
Gambini et al. [17]	TweepFake	Semantic Embeddings	✓
Tesfagergish et al. [18]	TweepFake	Semantic Embeddings	✓
D. Dukic et al. [19]	PAN [22]	Semantic Embeddings, Tweet metadata, Emoji embeddings	✓
Heidari and Jones [20]	Cresci et al.'s [23]	Semantic Embeddings, Sentiment features	✓
Fröhling et al. [9]	datasets from [28] [24] [25]	Linguistic features	-
Hamida et al. [21]	Cresci et al.'s	Linguistic features	-
Dickerson et al. [22]	IEDS	Tweet Syntax, Sentiment, User Behavior, Network	-
Our proposed model	Enhanced TweepFake*	Semantic Embeddings, Sentiment features, Linguistic features, Emoji embeddings,	✓

Note: TweepFake* denotes the newly Enhanced TweepFake dataset used in this work. NLM denotes complex neural language models.

III. Research Methodology

A. Dataset

To construct a comprehensive dataset of tweets generated by bots or humans, we adopted the TweepFake dataset [5]. The original TweepFake dataset contains five columns: user ID,

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status ID, screen name, account type, and class type (human, bot). Since the published dataset does not include the textual content of the tweets, we performed data scraping based on the provided status ID on Twitter using the Python library stweet [13]. However, approximately 30% of the tweets were no longer accessible due to tweet deletion or account deactivation. To address this limitation, we undertook additional searches to identify new bot accounts and gather supplementary tweet data, augmenting the original dataset. Acknowledging the emergence of advanced text generative models, such as GPT-3 and ChatGPT, we targeted Twitter bot accounts utilizing these latest generative models. This approach ensured the inclusion of up-to-date examples in the dataset. Consequently, we enriched the TweepFake dataset by incorporating tweets generated by the newest text-generative models alongside their corresponding human-authored tweets, creating an enhanced TweepFake dataset. An overview of the distribution of tweets in the enhanced dataset is in Table II.

TABLE II. OVERVIEW OF ENHANCED TWEEPFAKE DATASET

Account Type	Accounts	Tweets
Bot	24	10865
Human	17	10865
Total	41	21730

B. Feature Engineering

We obtained a matrix of feature values for training an effective classifier. To achieve this, we employed a feature engineering approach involving concatenating four distinct types of feature vectors: semantic features, emoji features, linguistic features, and sentiment features.

Semantic Features: we followed a two-step approach to capture the semantic features of the tweet text. Firstly, we performed fine-tuning on BERT pre-trained transformer model. By training the model on our labeled dataset, we allowed it to adapt to the nuances of our data. Secondly, we utilized the fine-tuned model's weights on the bare BERT model [26] from the Transformers Python library to extract word embeddings. These embeddings served as vectorized representations for each tweet, effectively capturing its contextual information and providing a solid foundation for subsequent classification tasks. Before proceeding with the two-step approach, we preprocessed the text data by replacing URLs with a <url> tag and mentions with a <user> tag. This step aimed to reduce noise and generalize the textual content. Subsequently, we transformed our dataset into a format suitable for training with BERT. The text data was tokenized using a BertTokenizer [30], ensuring a fixed sequence length of tokens through truncation and padding.

Emoji Features: to incorporate meaningful emoji features, we used emoji2vec [31], a pre-trained embedding model that assigns 300-dimensional vectors to all Unicode emojis based on their descriptions in the Unicode emoji standard. These emoji2vec embeddings provided valuable contextualized representations of emojis, which were crucial for our proposed discriminator. By incorporating the emoji2vec embeddings, we enriched our feature vector with informative emoji representations, enabling our classifier to leverage emojis' inherent meaning and nuances in a text.

Sentiment Features: to capture the sentiment expressed in the tweet text, we incorporated sentiment features using a pre-trained RoBERTaBASE sentiment classification model [32]. The sentiment classification model provides sentiment polarity scores for each tweet, indicating the strength of positive, negative, and neutral sentiment. These sentiment polarity features add an essential dimension to our feature vector, enabling our classifier to consider the emotional tone and sentiment conveyed by the text. By incorporating sentiment features, we aim to enhance the discriminative power of our classifier.

Linguistic Features: to capture the linguistic characteristics, we carefully selected 15 linguistic features based on their relative importance, as shown in Table III.

TABLE III. LINGUISTIC FEATURES DESCRIPTION

Features	Description
Tweet Indicators	
URL	Indicator of whether **URLs** present within the tweet.
Mentions	Indicator of whether **user mentions** present within the tweet.
Part of Speech (POS)	
NOUN	Ratio of **nouns** to the total number of tokens in a tweet.
VERB	Ratio of **verb** to the total number of tokens in a tweet.
PRON	Ratio of **pronoun** to the total number of tokens in a tweet.
PUNCT	Ratio of **punctuation** to the total number of tokens in a tweet.
PROPN	Ratio of **proper noun** to the total number of tokens in a tweet.
ADP	Ratio of **adposition** to the total number of tokens in a tweet.
DET	Ratio of **determiner** to the total number of tokens in a tweet.
ADJ	Ratio of **adjective** to the total number of tokens in a tweet.
AUX	Ratio of **auxiliary** to the total number of tokens in a tweet.
PART	Ratio of **particle** to the total number of tokens in a tweet.
CCONJ	Ratio of **coordinating conjunction** to a tweet total tokens.
Perplexity	
PPL	**Perplexity Score** based on GPT2 model

We employ binary indicators to identify the presence of URLs, and user mentions in the text data. We extracted POS features using spaCy linguistics analysis tool. Before feature extraction, tags such as <url>, <user>, and <hashtag> were removed to ensure accurate POS identification. The spaCy model, was then applied to the preprocessed text data. The relative frequency of each POS tag was calculated by dividing the count of each tag by the total number of tokens in the text data. To assess the complexity and predictive performance of the text samples in our dataset, we utilized the GPT-2 variant of the transformer-based language model [27] from the Python Transformers library. Perplexity scores were calculated by tokenizing the cleaned text and inputting it into the GPT-2 model. The model predicts each word in the sequence based on the preceding context. We obtained perplexity scores by

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comparing the predicted words with the actual words. In addition to replacing URLs with a <url> tag and user mentions with a <user> tag, we replaced hashtags with a <hashtag> tag followed by the hashtag content. These preprocessing steps help standardize and prepare the text data for accurate linguistic analysis and feature extraction.

C. Model Description

The proposed model architecture has two levels: feature extractor and classifier head. Fig. 1 provides a visual representation of the model schema, illustrating the integration of different features and classifiers. Our experiments explored feature combinations, including BERT embeddings alone and varieties of BERT embeddings with emoji, sentiment, and linguistic features. This approach allows us to leverage the strengths of each feature type and capture a comprehensive representation of the input data. To construct the feature vector for each training example, the feature above vectors are concatenated, creating a unified representation that serves as the input for our classifier. We opted for a Multi-Layer Perceptron (MLP) to incorporate additional features beyond BERT.

In addition to the MLP, we evaluated our features with shallow ML models, including logistic regression (LR) [27], support vector machines (SVC) [28], and random forest (RF) [29]. The inference for using these models is that it provides an initial benchmark for assessing the performance of different classifiers and identifying the optimal approach for MGT detection. To implement the baseline models using standard ML methods, we utilized scikit-learn [30]. This widely-used ML toolkit provides an extensive collection of algorithms and utilities, making it a valuable resource for implementing and evaluating our models. We leveraged the Keras library [32] for the DL models for its ease of use and flexibility.

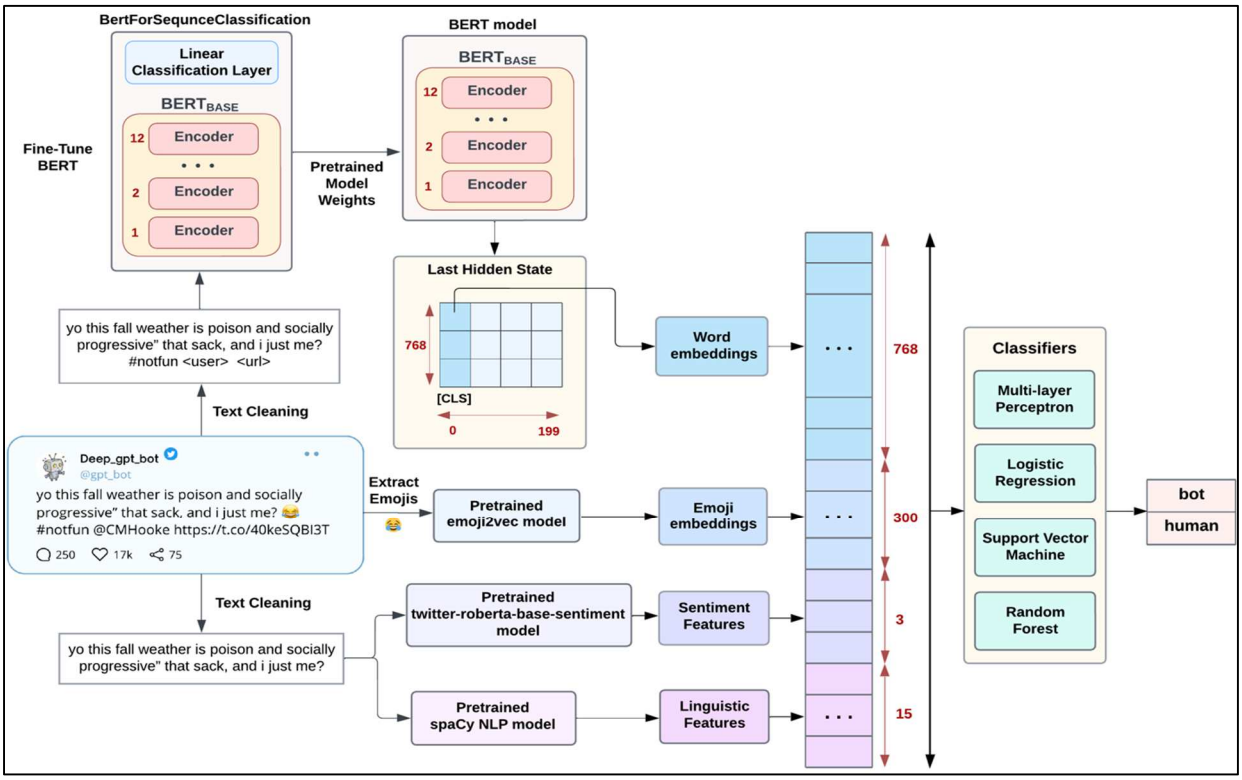

Fig. 1. A model schema for the proposed set of features.

IV. Results and Discussion

Our analysis revealed several distinguishing characteristics between MGT and HWT, shedding light on their unique properties and behaviors. One prominent difference was observed in tweet indicators, user mentions, and URLs, where HWT displayed a higher frequency than MGT (Fig. 2). This finding presents that human users are likely more inclined to engage with others by mentioning them and sharing external links in their tweets. In contrast, bots generate content with fewer mentions and less sharing of external links, indicating a lack of genuine connections and social interactions on Twitter. Furthermore, we observed no noticeable difference in the usage of hashtags between the two account types, indicating both MGT and HWT utilize hashtags similarly, potentially for organizing topics or enhancing discoverability. These distinguishing characteristics shed light on the unique dynamics and underlying behaviors of MGT and HWT, emphasizing the central role of human social interaction and information sharing in the creation of authentic content on Twitter.

The POS analysis findings show that MGT tends to utilize a higher frequency of words related to NOUN (noun), PRON (pronoun), DET (determiner), and ADP (adposition) compared to HWT (Fig. 3). Conversely, MGT employs fewer words related to PROPN (proper noun) and ADV (adverb). This finding shows that MGT generative models are proficient at replicating common language patterns involving these parts of

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speech. They learn from large text corpora, enabling them to replicate the grammatical structures commonly associated with nouns, pronouns, determiners, and adpositions. However, these models may struggle with capturing the nuances and context of proper nouns and adverbs, leading to their lower usage in MGT.

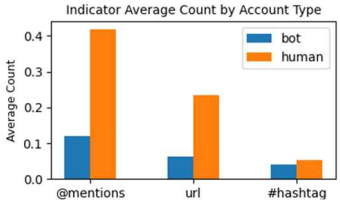

Fig. 2. Tweet Indicators Distribution grouped by Account Types.

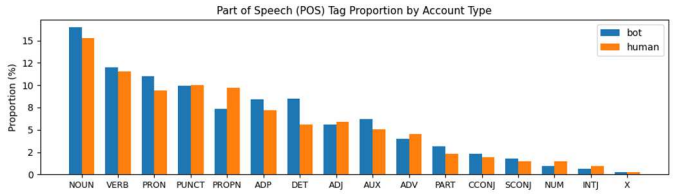

Fig. 3. Part-of-speech distribution grouped by Account Type.

In analyzing named entities, MGT demonstrates a higher frequency of words related to PERSON entities, possibly due to the models' exposure to social media and online content where individuals' names and mentions are prevalent. Conversely, MGT exhibits a lower frequency of words related to DATE entities, presenting less time-dependent content and more focused on general patterns and topics (Fig. 4). This finding indicates that generative models prioritize generating applicable and relevant content beyond specific dates or temporal contexts.

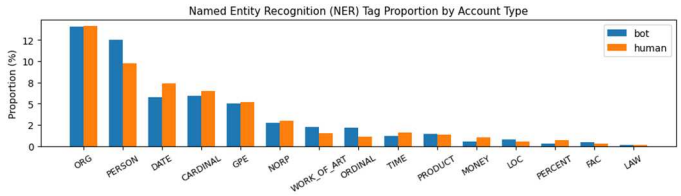

Fig. 4. Named Entity Distributions grouped by Account Type.

The text perplexity scores revealed a bimodal distribution in MGT, exhibiting two distinct peaks (Fig. 5, left). We performed a post-hoc analysis (Fig. 5, right), confirming that modern MGT (such as GPT-2, GPT-3, and ChatGPT) exhibits relatively low perplexity compared to traditional MGT (such as RNN). This result is displayed likely because the traditional models may have higher perplexity due to their limited capacity to capture the intricacies of language and context. Regarding linguistic feature importance, Perplexity emerges as the most crucial variable, followed by the '@mentions' indicator and multiple POS tags, indicating their strong discriminatory power (Fig. 6). On the other hand, Name Entity Recognition (NER) features and the hashtag show comparatively lower relative importance in the classification task. These finding hints that language complexity and unpredictability, user engagement indicators, and grammatical patterns play vital roles in distinguishing between MGT and HWT. Lastly, sentiment analysis reveals differences in sentiment patterns between MGT and HWT. MGT exhibits a relatively higher proportion of neutral sentiment but a lower proportion of positive sentiment than HWT (Fig. 7), demonstrating that generative models used for MGT tend to produce content with a neutral tone, avoiding strong opinions or emotions. This tendency aligns with their objective of maintaining a more impersonal and non-controversial stance.

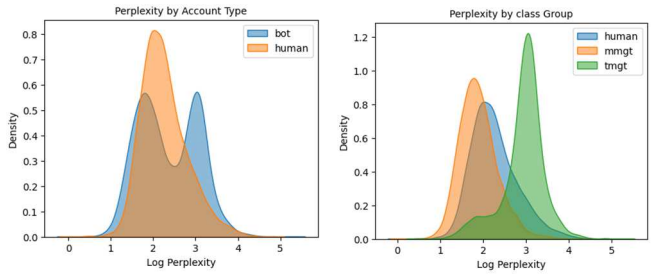

Fig. 5. Perplexity Distribution of human-written text and machine-generated text by Account Type.

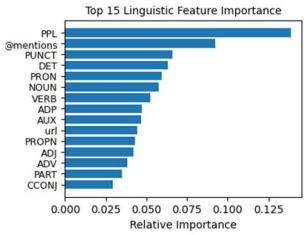

Fig. 6. Linguistic Feature Importance.

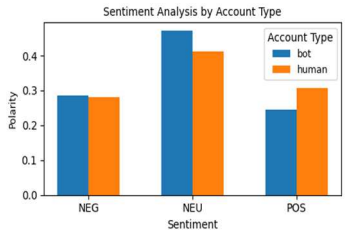

Fig. 7. Sentiment Distribution

Overall, the analysis of distinguishing characteristics of MGT and HWT on Twitter reveals that HWT tends to engage more with others and share external content through user mentions and URLs. Conversely, MGT demonstrates proficiency in replicating common language patterns associated with nouns, pronouns, determiners, and adpositions. MGT also prefers generating content focused on individuals rather than specific dates or temporal contexts. Moreover, MGT shows a higher proportion of neutral sentiment compared to HWT.

TABLE IV. COMPARISON OF RESULTS USING ENHANCED TWEEPFAKE

Related Research	Features of Best model	Model Performance
Fagni et al. 2021 [17]	Fine-tuned RoBERTa model	Acc: 87.6% $F1_{Bot}$: 87.6% $F1_{Human}$: 87.6%
Our Proposed Model	Fine-tuned BERT (word embeddings) + emoji2vec embeddings + linguistic features + Multi-Layer Perceptron (classifier)	Acc: 88.3% $F1_{Bot}$: 88.3% $F1_{Human}$: 88.2%

To provide a reference for our model performance, we compared our results with existing literature that utilized the TweepFake dataset, specifically the work by Fagni et al. [5] (Table IV). However, direct comparisons between our research and Fagni's work, as well as other studies using the same dataset, are not directly applicable due to the dissimilarity between our Enhanced TweepFake dataset and the original TweepFake dataset. Therefore, the reported performance should be considered in the context of this dataset limitation. Our model's

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performance achieves accuracy scores of 88.3% and F1 scores ranging from 88.1% to 88.3%.

V. CONCLUSION

Our research unveiled apparent disparities between MGT and HWT. These findings highlight the necessity of considering these nuances when designing classification models. We also identified several future research directions. Firstly, there is potential for exploring other variations of BERT models, such as RoBERTa and DistilBERT, as well as incorporating models like XLNET, which have exhibited promising results in related studies [2], to enhance the model's performance. Additionally, we propose incorporating more sophisticated linguistic features and conducting in-depth feature importance analysis, such as utilizing SHAP (SHapley Additive exPlanations), to gain deeper insights into the discriminative power of individual features.

REFERENCES

[1] "Introducing ChatGPT," OpenAI, Nov. 30, 2022. https://openai.com/blog/chatgpt (accessed Jul. 17, 2023).

[2] E. Ferrara, O. Varol, C. Davis, F. Menczer, and A. Flammini, "The rise of social bots," Commun. ACM, vol. 59, no. 7, pp. 96–104, Jun. 2016.

[3] Q. Khan and H.N. Chua, "Comparing topic modeling techniques for identifying informative and uninformative content: A case study on COVID-19 tweets," in 2021 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET), September 2021, pp. 1-6, IEEE.

[4] Jan Hendrik Kirchner, Lama Ahmad, Scott Aaronson, and Jan Leike, "New AI classifier for indicating AI-written text," OpenAI, Jan. 31, 2023.

[5] T. Fagni, F. Falchi, M. Gambini, A. Martella, and M. Tesconi, "TweepFake: About detecting deepfake tweets," PLOS ONE, vol. 16, no. 5, p. e0251415, May 2021.

[6] H. Allcott and M. Gentzkow, "Social Media and Fake News in the 2016 Election," J. Econ. Perspect., vol. 31, pp. 211–236, May 2017.

[7] H. Allcott and M. Gentzkow, "Social Media and Fake News in the 2016 Election," J. Econ. Perspect., vol. 31, pp. 211–236, May 2017.

[8] A. Vaswani et al., "Attention Is All You Need." Advances in neural information processing systems, 2017, 30.

[9] L. Fröhling and A. Zubiaga, "Feature-based detection of automated language models: tackling GPT-2, GPT-3 and Grover," PeerJ Comput. Sci., vol. 7, p. e443, Apr. 2021.

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.", arXiv preprint arXiv:1908.04577 (2019).

[11] A. Radford, J. Wu, R. Child, D. Luan, D. Amodel, and I. Sutskever, "Language Models are Unsupervised Multitask Learners," OpenAI, Feb. 14, 2019.

[12] Solaiman et al., "Release Strategies and the Social Impacts of Language Models." arXiv, Nov. 12, 2019.

[13] T. B. Brown et al., "Language Models are Few-Shot Learners." arXiv, Jul. 22, 2020. .

[14] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach." arXiv, Jul. 26, 2019.

[15] E. Crothers, N. Japkowicz, and H. Viktor, "Machine Generated Text: A Comprehensive Survey of Threat Models and Detection Methods." arXiv, Oct. 13, 2022.

[16] S. M. Saravani, I. Ray, and I. Ray, "Automated Identification of Social Media Bots Using Deepfake Text Detection," in Information Systems Security, S. Tripathy, R. K. Shyamasundar, and R. Ranjan, Eds., in Lecture Notes in Computer Science, vol. 13146. Cham: Springer International Publishing, 2021, pp. 111–123.

[17] M. Gambini, D. M. Tesconi, and D. F. Falchi, "Developing and experimenting approaches for DeepFake text detection on Social Media," University of Pisa, 2020. [Online]. Available: https://etd.adm.unipi.it/t/etd-07032020-115029/.

[18] S. G. Tesfagergish, R. Damaševičius, and J. Kapočiūtė-Dzikienė, "Deep Fake Recognition in Tweets Using Text Augmentation, Word Embeddings and Deep Learning," in Computational Science and Its Applications – ICCSA 2021: 21st International Conference, Cagliari, Italy, September 13–16, 2021, Proceedings, Part VI, Berlin, Heidelberg: Springer-Verlag, Sep. 2021, pp. 523–538.

[19] [48] D. Dukic, D. Keca, and D. Stipic, "Are You Human? Detecting Bots on Twitter Using BERT," in 2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA), sydney, Australia: IEEE, Oct. 2020, pp. 631–636.

[20] M. Heidari and J. H. Jones, "Using BERT to Extract Topic-Independent Sentiment Features for Social Media Bot Detection," in 2020 11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), Oct. 2020, pp. 0542–0547.

[21] Z. F. Hamida, A. Refouf, A. Drif, and S. Giordano, "Hybrid-MELAu: A Hybrid Mixing Engineered Linguistic Features Framework Based on Autoencoder for Social Bot Detection.," Informatica, vol. 46, no. 6, Art. no. 6, Sep. 2022.

[22] J. P. Dickerson, V. Kagan, and V. S. Subrahmanian, "Using sentiment to detect bots on Twitter: Are humans more opinionated than bots?," in 2014 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2014), Aug. 2014, pp. 620–627.

[22] F. Rangel and P. Rosso, "Overview of the 7th Author Profiling Task at PAN 2019: Bots and Gender Profiling in Twitter," p. 36.

[23] S. Cresci, R. Di Pietro, M. Petrocchi, A. Spognardi, and M. Tesconi, "The Paradigm-Shift of Social Spambots: Evidence, Theories, and Tools for the Arms Race," in Proceedings of the 26th International Conference on World Wide Web Companion - WWW '17 Companion, Perth, Australia: ACM Press, 2017, pp. 963–972.

[24] R. Zellers et al., "Defending Against Neural Fake News," p. 12.

[25] J. Clark, A. Radford, and J. Wu, "Gpt-2 simple baseline." 2019. [Online]. Available: https://github.com/openai/ gpt-2-output-dataset/blob/master/detection.

[26] OpenAI, "GPT-4 Technical Report." arXiv, Mar. 27, 2023. doi: 10.48550/arXiv.2303.08774.

[27] A. Radford, J. Wu, R. Child, D. Luan, D. Amodel, and I. Sutskever, "Language Models are Unsupervised Multitask Learners," OpenAI, Feb. 14, 2019. https://openai.com/blog/better-language-models/ (accessed Nov. 10, 2022).

[28] T. B. Brown et al., "Language Models are Few-Shot Learners." arXiv, Jul. 22, 2020. doi: 10.48550/arXiv.2005.14165.

[29] OpenAI, "GPT-4 Technical Report." arXiv, Mar. 27, 2023. doi: 10.48550/arXiv.2303.08774.

[30] "Introducing ChatGPT," OpenAI, Nov. 30, 2022. https://openai.com/blog/chatgpt (accessed Jul. 17, 2023).

[31] B. Eisner, T. Rocktäschel, I. Augenstein, M. Bošnjak, and S. Riedel, "emoji2vec: Learning Emoji Representations from their Description." arXiv, Nov. 20, 2016.

[32] A. Fan, M. Lewis, and Y. Dauphin, "Hierarchical Neural Story Generation," in Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 889–898. doi: 10.18653/v1/P18-1082.

[33] C. J. Lee and H. N. Chua, "Using Linguistics and Psycholinguistics Features in Machine Learning for Fake News Classification Through Twitter", Proceedings of International Conference on Data Science and Applications, pp. 717-730, 2021.

[34] Q. Khan and H.N. Chua, "An Automated Topics Labelling Framework using Zero-Shot Text Classification," in Journal of Engineering Science and Technology, Special Issue on ACSAT, pp. 46-59, 2021, School of Engineering, Taylor's University.

[35] H. N. Chua, Q. Khan, M. B. Jasser, and R. T. Wong, "Problem Understanding of Fake News Detection from a Data Mining Perspective," in 2023 IEEE 13th International Conference on Control System, Computing and Engineering (ICCSCE), 2023, pp. 297-302.

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